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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Machine Learning (course)

Week 9: Assignment 9 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) Consider the bayesian network shown below. ***1 point***

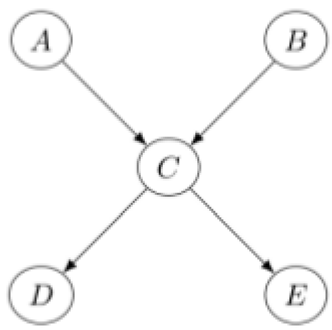

Figure 1: Q1

Two students - Student A and Student B make the following claims:

- Student A claims $P(A|\{B, D\}) = P(A|D)$
- Student B claims $P(A|B) = P(A)$

where $P(X|Y)$ denotes probability of event X given Y . Please note that Y can be a set. Which of the following is true?

- Student A and Student B are correct.
- Student A is correct and Student B is incorrect.
- Student A is incorrect and Student B is correct.
- Both are incorrect.

Insufficient information to make any conclusion. Probability distributions of each variable should be given.

2) Consider the Bayesian graph shown below in Figure 2. ***1 point***

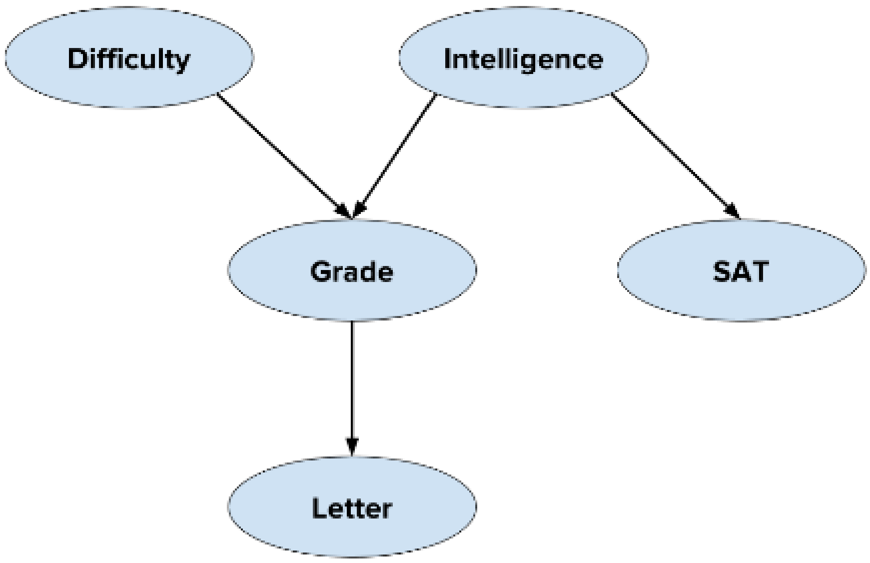

Figure 2: Q3

The random variables are modeled as discrete variables and the corresponding CPDs are as below.

d^0	d^1
0.6	0.4

i^0	i^1
0.6	0.4

	g^1	g^2	g^3
i^0, d^0	0.3	0.4	0.3
i^0, d^1	0.05	0.25	0.7
i^1, d^0	0.9	0.08	0.02
i^1, d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.2	0.8
g^2	0.4	0.6
g^3	0.99	0.01

What is the probability of $P(i = 1, d = 0, g = 2, s = 1, l = 0)$?

0.004608

0.006144

- Quiz: Week 9 : Assignment 9 (assessment?name=301)
- **Week 10 ()**
- **Week 11 ()**
- **Week 12 ()**
- **Text Transcripts ()**
- **Download Videos ()**
- **Books ()**
- **Problem Solving Session - July 2024 ()**

○ 0.003992
○ 0.007309
○ None of these

3) Consider the Markov network shown below in Figure 3 ***1 point***

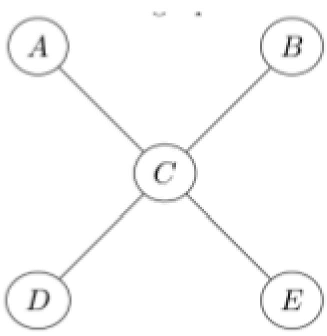

Figure 3: Q4

Two students - Student A and Student B make the following claims:

- Student A claims $P(A|B) = P(A)$
- Student B claims $P(A|\{B, C\}) = P(A|C)$

Which of the following is true?

○ Student A and Student B are correct.
○ Student A is incorrect and Student B is correct.
○ Student A is correct and Student B is incorrect.
○ Both are incorrect.
○ Insufficient information to make any conclusion. Probability distributions of each variable should be given.

4) In the Markov network given in Figure 4, two students make the following claims: ***1 point***

- Manish claims variable “1” is independent of variable “6” given variable “3”.
- Trina claims variable “1” is independent of variable “8” given variable “4”.

Which of the following is true?

○ Both the students are correct.
○ Trina is incorrect and Manish is correct.
○ Trina is correct and Manish is incorrect.
○ Both the students are incorrect.
○ Insufficient information to make any conclusion. Probability distributions of each variable should be given.

5) Does there exist a more compact factorization involving less number of factors for the distri- bution given in Equation 1 in previous question? ***1 point***

○ Yes
○ No

○ Insufficient information

Check Answers and Submit